

Robust Modulation Classification Over α -Stable Noise Using Graph-Based Fractional Lower-Order Cyclic Spectrum Analysis

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Abstract—This paper introduces a novel automatic modulation classification (AMC) method using the graph-based fractional lower-order cyclic-spectrum analysis in the α -stable noise environment. The noise in the practical communication scenario usually exhibits impulse characteristics in the statistical sense, which could be modeled as the α -stable distribution. This would make the second- or higher-order statistics of the received signal vanish, and thus the performances of the conventional AMC algorithms designed for Gaussian noise significantly deteriorate when directly employed in the α -stable noise. In our proposed framework, the fractional lower-order cyclic spectrum (FLOCS) analysis is first invoked to acquire the polyspectra of the signal corrupted by the α -stable noise. Then, the graph-based AMC mechanism is systematically established upon the graph representation of the FLOCS to identify the modulation type according to the discrepancies between the graph features derived from the training and test data. The performance of our proposed new algorithm is theoretically analyzed, and the correct classification probability P_{cc} over the modulation candidate set is formulated analytically. The remarkable scalability and efficiency of our proposed approach for the modulation candidate set variation are also theoretically investigated in detail. Monte Carlo simulation results demonstrate the effectiveness and superiority of the proposed AMC scheme.

Index Terms—Modulation classification, α -stable distribution, fractional lower-order cyclic spectrum, graph representation, adjacency matrix, Kullback-Leibler divergence, Hamming distance.

I. INTRODUCTION

AUTOMATIC modulation classification (AMC), which is regarded as an essential and intricate operation between signal detection and demodulation in an intelligent receiver, can autonomously identify the modulation type of an unknown

received signal, which would be corrupted by noise and fading channels. AMC becomes a rapidly evolving area with various civilian and military applications such as cognitive radio, adaptive modulation, *intelligent transportation systems* (ITSs), surveillance and electronic warfare [1]–[3]. It is considered as a promising and indispensable technique to enhance the spectrum sensing capability of intelligent vehicular communications in ITSs, such as *vehicle-to-vehicle* (V2V) and *vehicle-to-infrastructure* (V2I) wireless communications [4].

The traditional AMC techniques, which can generally divided into two categories: (i) the *likelihood-based* (LB) decision-theoretic approach (see [5], [6]) and (ii) the *feature-based* (FB) approach (see [7]–[11]), have almost all been investigated under the assumption of Gaussian noise [12]–[14]. However, various studies have shown that the interference in the practical wireless communication channels usually appears to have notable impulses caused by multiple access, low-frequency atmospheric noise, electromagnetic disturbance, etc. [15]–[18]. These physical noises (disturbances), due to a variety of natural and man-made sources, exhibit sharp spikes and their density functions decay in the tails less rapidly than Gaussian density function. This *non-Gaussian* noise, which presents impulse characteristics in the statistical sense, is actually one of the primary sources of error in any wireless communication system, and can be effectively modeled as an α -stable distribution according to the *Generalized Central Limit Theorem* [19]–[23]. It must be noticed that α -stable distribution does not have a closed-form probability density function [19], and is simplified as a Gaussian distribution when its *characteristic exponent* α is equal to 2. This crucial property makes the second- and higher-order statistics of the signal corrupted by α -stable noise vanish. Thus, the conventional AMC approaches, which are designed to achieve an analytical optimal solution under the Gaussian assumption, become invalid when the α -stable noise is present. Note that our previous AMC methods in [24], [25] can only deal with Gaussian noise. The higher-order statistics based AMC method in [24] must rely on a very large sample size while the graph-based AMC scheme in [25] is built upon simple *constellation-angles*, which could not be reliable for non-Gaussian noise.

AMC in the α -stable noise has been investigated in recent years: The *sparse signal decomposition* (SSD) based *maximum likelihood* (ML) classification method was devised for AMC in the presence of α -stable noise [26]. This LB method utilized

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an overcomplete dictionary to capture morphological components in deterministic modulated signals and further identify the modulation type using ML; nevertheless, this scheme worked only for low-order M -ary phase-shift keying (M -PSK) and M -ary quadrature amplitude modulation (M -QAM) signals satisfactorily. The Kolmogorov-Smirnov (K-S) test is invoked to identify the modulation schemes of M -PSK and M -QAM signals according to the empirical cumulative distribution function (ECDF) of some decision statistics generated from the received signals [27]. Unfortunately, the K-S classifier often leads to poor performance in the low mixed signal-to-noise ratio (MSNR) regime; the multifractal spectrum characteristics of the received M -ary frequency-shift keying (M -FSK) signals corrupted by α -stable noise are used as the modulation features to identify their modulation orders [28]. However, this scheme is only valid for M -FSK signals; in [29], the modulation features are constructed by the lower-order wavelet packet decomposition, and the radial basis function (RBF) neural network is utilized to classify digital modulation signals. This method can tackle a broader modulation candidate set, but the neural network classifier in [29] leads to the tremendous computational workload. The fractional lower-order statistics (FLOS) based fourth-order cumulants serve as the key features for classification, and the modulation schemes are recognized by comparing the features with theoretical thresholds [30]. The simulation results show that this method results in promising performance but suffers from high computational complexity. Thus, the existing techniques for AMC in the presence of α -stable noise are either computationally cumbersome or leading to unsatisfactory performances. Consequently, new robust and efficient modulation classification method(s) for the α -stable noise background are in high demand.

Recently, a novel automatic modulation classifier using the graph-based cyclic-spectrum analysis, namely AMC_G , has just emerged and lead to promising performance [31], [32]. Since the second-order statistics of the modulated signal corrupted by α -stable noise vanish, this AMC_G method is theoretically inapplicable for the AMC in the α -stable noise. Nevertheless, AMC_G , which is the first attempt to introduce AMC in the graph domain, provides a new idea for AMC design over the α -stable noise. On the other hand, it must be noticed that the α -stable distribution indeed has the finite lower-order statistics, where the order is less than 2. Hence, the fractional lower-order cyclic spectrum (FLOCS) analysis can be employed to analyze the signal degraded by α -stable noise in the polyspectra domain.

In this work, the graph-based AMC approach is systematically built on the FLOCS graph representation of the modulated signal corrupted by α -stable noise. The FLOCS of the received signal corrupted by α -stable noise is first obtained by the transformation of its fractional lower order moments. The distinguishable features are iteratively extracted from the adjacency matrices corresponding to the graph representation of FLOCS. According to the normalized Hamming distance, which measures the discrepancies between the features derived from the training and test data, the modulation classification can be performed. In order to exhibit the promising scalability and efficiency of our proposed method, the novel iterative modulation feature-selection mechanism is established for the modulation candidate

set variation (*expansion* or *reduction*). Then, the performance of the proposed AMC method is investigated in detail, and the correct classification probability P_{cc} over the modulation candidate set is derived *analytically*. Through the theoretical analysis, it is apparent that the iterative feature-selection scheme based on the Kullback-Leibler divergence in our algorithm avoids any human involvement, and can be extended to any arbitrary modulation candidate set systematically. When the modulation candidate set is expanded or shrunk, only the features for the modulation candidate(s) which are added to or removed from the original modulation candidate set need to be calculated, and P_{cc} for the new scaled modulation candidate set can be obtained by a minor correction of the original correct classification probability. Since only the correction term(s) resulting from the added/removed modulation candidate(s) need to be calculated, the computation complexity is significantly reduced. Finally, the effectiveness and superiority of our proposed AMC approach are justified by Monte Carlo simulations. Note that the *non-uniform spaced constellations* adopted in the recent ATSC standard for modern digital television (see [33]–[35]) will not be considered in this work.

The rest of the paper is organized as follows: Section II introduces the backgrounds of α -stable noise and fractional lower-order cyclic spectrum, which are necessary for deriving our new AMC method. In Section III, our proposed new AMC technique over the α -stable noise and the associated iterative modulation feature-selection mechanism are discussed in detail. The performance of our proposed method is then theoretically analyzed, and P_{cc} over the modulation candidate set is formulated in Section IV. The effectiveness and superiority of our proposed new AMC scheme are justified by Monte Carlo simulations in Section V. Finally, conclusion will be drawn in Section VI.

II. BACKGROUND

In this section, the α -stable distribution is first introduced. Then, the FLOCS of the signal corrupted by α -stable noise is investigated, and the effectiveness of the FLOCS analysis for such contaminated communication signal is also demonstrated.

A. α -Stable Distribution

The α -stable distribution, also known as non-Gaussian random distribution or heavy-tailed distribution, is regarded as a kind of *generalized Gaussian distribution*, which can be invoked to accurately describe the statistic characteristics of the noise emerging in the practical wireless communication environment [21]. The α -stable distribution satisfies the generalized central limit theorem but does not have a closed-form probability density function [19]. However, the random variable u , which represents an α -stable process, can be defined by the *unified characteristic function* (UCF) given by

$$\phi(u) \stackrel{\text{def}}{=} \exp \left\{ j\alpha u - \gamma |u|^\alpha [1 + \beta \text{sgn}(u)\omega(u, \alpha)] \right\}, \quad (1)$$

where

$$\omega(u, \alpha) = \begin{cases} \tan\left(\frac{\pi\alpha}{2}\right), & \alpha \neq 1, \\ \left(\frac{2}{\pi}\right) \ln|u|, & \alpha = 1, \end{cases} \quad (2)$$

$j \stackrel{\text{def}}{=} \sqrt{-1}$, $\ln(\cdot)$ is the natural logarithm operation, and $\text{sgn}(\cdot)$ is the *sign* function, which is defined by

$$\text{sgn}(u) \stackrel{\text{def}}{=} \begin{cases} 1, & u > 0, \\ 0, & u = 0, \\ -1, & u < 0. \end{cases} \quad (3)$$

Note that α denotes the *characteristic exponent*, which illustrates the impulsive characteristics of the distribution and controls the *heaviness* of the tails ($0 < \alpha \leq 2$). The smaller the value of α , the thicker (heavier) the tail of the corresponding distribution (it leads to more substantial impulsive characteristics). In addition, β is the *skewness parameter* which characterizes the degree of symmetry of the distribution ($-1 \leq \beta \leq 1$); γ is referred to as the *scale parameter* or *dispersion*, which is similar to the variance of a Gaussian distribution; a is the mean of the α -stable distribution. It must be noticed that the α -stable distribution can be regarded as a generalization of Gaussian distribution with the following properties:

- when $\alpha = 2$, the α -stable distribution is simplified as a Gaussian distribution;
- when $\alpha = 1$ and $\beta = 0$, the α -stable distribution is simplified as a Cauchy distribution;
- when $\beta = 0$, the α -stable distribution is symmetric with respect to its mean a and called the S α S distribution.

In our work, we focus on the zero-mean ($a = 0$) S α S distribution with which the noise in the practical wireless communication environment complies.

B. Fractional Lower-Order Cyclic Spectrum

Assume the modulated signal $s(t)$ is contaminated by the additive α -stable noise $v(t)$ via the transmission channel, and the received signal $x(t)$ can be expressed by

$$x(t) = s(t) + v(t). \quad (4)$$

Here, $v(t)$ follows the symmetric α -stable distribution, or the S α S distribution. The received signal $x(t)$ is first uniformly sampled at the sample frequency $F_s = 1/T_s$. It is well known that the second- or higher-order statistics of the modulated signal which is corrupted by α -stable noise do not exist [22], and the MSNR can be invoked to describe the signal-to-noise condition of α -stable noise, which is defined by

$$\text{MSNR} \stackrel{\text{def}}{=} 10 \log_{10} (\sigma_s^2 / \gamma), \quad (5)$$

where σ_s^2 is the variance of $s(t)$, and $\log_{10}(\cdot)$ denotes the logarithm operation of base 10. It is apparent from Figs. 1(a), 1(c), and 1(e) that the *second-order cyclic spectra* (SCS's) of the 2FSK, 4FSK, and BPSK signals corrupted by α -stable noise are completely submerged by the noise floor, and thus the distinguishable spectral features of those signals cannot be found. Fortunately, the FLOCS, which is generated from the nonlinear transformation of the signal, is widely considered as an effective

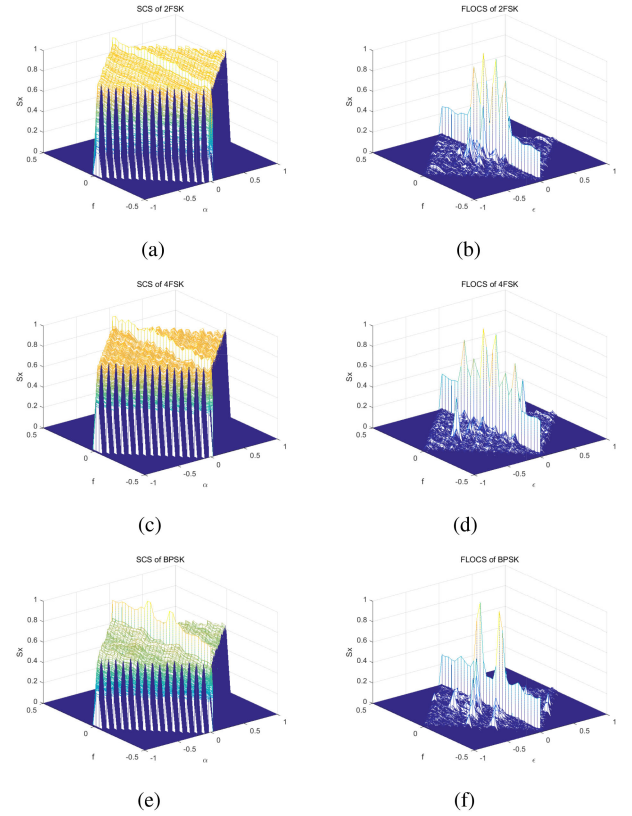


Fig. 1. The second-order cyclic spectra and the FLOCS's of some typical modulated signals corrupted by α -stable noise (MSNR = 0 dB). (a) The SCS of 2FSK. (b) The FLOCS of 2FSK. (c) The SCS of 4FSK. (d) The FLOCS of 4FSK. (e) The SCS of BPSK. (f) The FLOCS of BPSK.

tool to analyze the signal degraded by α -stable noise in the polyspectra domain.

The *fractional lower-order cyclic correlation* (FLOCC) of the received discrete-time signal sequence $x(n)$ can be calculated by

$$\text{FLOCC}(n, \tau) \stackrel{\text{def}}{=} E \left\{ [x(n + \tau)]^{\{b\}} [x^*(n)]^{\{b\}} \right\}, \quad (6)$$

where

$$x(n)^{\{b\}} \stackrel{\text{def}}{=} |x(n)|^{b-1} x^*(n), \quad (7)$$

and Eq. (7) is the b^{th} order nonlinear transformation of $x(n)$, $0 < b < \alpha/2$; $E\{\cdot\}$ is statistical-expectation operation and $x^*(n)$ is the complex conjugate of $x(n)$. Then, the FLOCC can be computed as

$$R_x^\varepsilon(\tau T_s) \stackrel{\text{def}}{=} \left\langle [x(nT_s + \tau T_s)]^{\{b\}} [x^*(nT_s)]^{\{b\}} e^{-j2\pi\varepsilon nT_s} \right\rangle \times e^{-j2\pi\varepsilon \tau T_s}, \quad (8)$$

where $\langle \cdot \rangle$ indicates the time-average operation. It must be noticed that the b^{th} order nonlinear transformation of $x(n)$ in Eq. (7) alters only the amplitudes of $x(n)$ and thus has no impact on its cyclic period. Hence, the definition of cyclic frequency ε in the traditional second-order cyclic spectrum analysis is also applicable for the FLOCS analysis. If $b = 1$, the FLOCC is directly reduced to the second-order cyclic correlation. Finally,

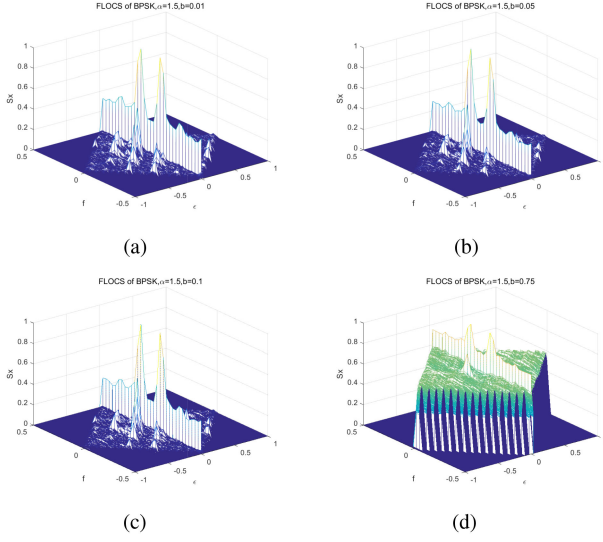


Fig. 2. The FLOCS's of the BPSK signal corrupted by α -stable noise with different b 's (MSNR = 0 dB, $\alpha = 1.5$). (a) $b = 0.01$. (b) $b = 0.05$. (c) $b = 0.1$. (d) $b = 0.75$.

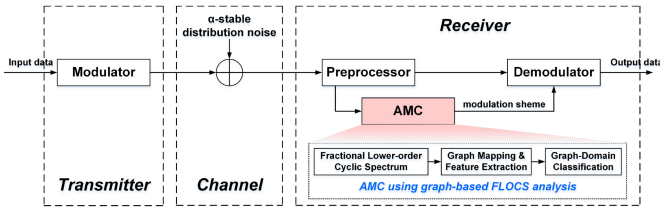


Fig. 3. The architecture of our proposed new AMC system.

the FLOCS, namely the Fourier transform of the FLOCC, can be obtained as

$$S_x^\varepsilon(f) \stackrel{\text{def}}{=} \sum_{h=-\infty}^{+\infty} R_x(hT_s) e^{-j2\pi f h T_s}, \quad (9)$$

where f is the signal frequency. The FLOCS's of 2FSK, 4FSK, and BPSK are depicted in Fig. 1(b), Fig. 1(d), and Fig. 1(f), respectively. Compared to the corresponding SCS's of those signals in Fig. 1(a), Fig. 1(c), and Fig. 1(e), it is obvious that noise floors are significantly mitigated by use of the FLOCS analysis, and the spectral features in FLOCS's can be extracted to differentiate those signals. Besides, it should also be noticed that the order of the nonlinear transformation in FLOCS, says b , affects the performance of the α -stable noise suppression. From Fig. 2, it is apparent that the α -stable noise floors in the FLOCS's of BPSK rise up with the increase of b . For the FLOCS analysis, b is empirically set to 0.1 for AMC.

III. OUR PROPOSED NEW AMC SCHEME

The architecture of our proposed new AMC system is depicted in Fig. 3. It is conspicuous that AMC, which is an indispensable mechanism between signal detection and demodulation, is composed by three major components: the *fractional lower-order cyclic spectrum analysis*, the *graph mapping & feature extraction mechanism*, and the *graph-domain classifier*. For

the “intelligent” receiver for non-cooperative communications, the received signal is first processed to eliminate the carrier frequency offset and the symbol timing offset via high-precision blind synchronization. Then, the modulation scheme of the received signal $x(t)$ is identified by the AMC scheme. Assume that the modulation candidate set $\mathbb{K}$ is given and it is defined by $\mathbb{K} \stackrel{\text{def}}{=} \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_K\}$, where $\mathcal{K}_k$ denotes the k^{th} modulation type, $k = 1, 2, \dots, K$.

A. Fractional Lower-Order Cyclic Spectrum Analysis

Since the moments of order greater than α for an α -stable process are unbounded, the traditional AMC algorithms based on cyclostationarity or *high-order statistics* (HOS) become inapplicable, when they are directly applied to the signal degraded by α -stable noise. However, it should also be noticed that the fractional lower-order statistics of the α -stable process exist (with bounded values) indeed. Thus, the FLOCS can be exploited to analyze the received signal in the polyspectra domain and transformed into the graph domain.

In practice, the FLOCS $S_x^\varepsilon(f)$ defined by Eq. (9) can be estimated from the nonlinear transformation of $x(n)$ by utilizing a time-smooth algorithm called the *fast Fourier transform* (FFT) *accumulation method* (FAM). The *time-smoothed cyclic periodogram* for a given signal frequency f and a cycle frequency ε is defined by

$$S_{x_T}^\varepsilon(n, f)_{\Delta t} \stackrel{\text{def}}{=} \sum_r X_T(r, f_1) X_T^*(r, f_2) g(n-r), \quad (10)$$

where $g(n)$ is a unity area weighting function of width $\Delta t = NT_s$ seconds, f_1 and f_2 are the center frequencies of the filters used in the FAM, i.e., $f_1 = f + \varepsilon/2$, $f_2 = f - \varepsilon/2$, and $X_T(r, f_1)$, $X_T(r, f_2)$ are the complex demodulates of $x(n)$'s nonlinear transform. That is,

$$X_T(n, f) \stackrel{\text{def}}{=} \sum_{r=-N'/2}^{N'/2-1} A(r) x(n-r)^{\{b\}} e^{-j2\pi f(n-r)T_s}, \quad (11)$$

where N' is the number of spectral frequencies; $A(r)$ is a data tapering window of duration $T = N'T_s$ seconds, and its bandwidth is also the frequency resolution Δf of the FLOCS. If $A(r)$ is normalized, the FLOCS can be estimated from the time-smoothed cyclic periodogram without bias such that

$$S_x^\varepsilon(f) \stackrel{\text{def}}{=} \lim_{\Delta f \rightarrow 0} \lim_{\Delta t \rightarrow \infty} S_{x_T}^\varepsilon(n, f)_{\Delta t}. \quad (12)$$

According to Eqs. (10)–(12), the FLOCS's of some typical modulated signals disturbed by α -stable noise are illustrated in Fig. 1(b), Fig. 1(d), and Fig. 1(f), respectively.

B. Graph Representation

By the FLOCS analysis, one can obtain a three-dimensional FLOCS $S_x^\varepsilon(f)$ with non-negative amplitudes, which contains $N' + 1$ spectral frequencies $f = f_v$, for $v = -N'/2, -N'/2 + 1, \dots, N'/2$, and $2N + 1$ cyclic frequencies $\varepsilon = \varepsilon_d$, for $d = -N, -N + 1, \dots, N$. The graph-mapping method presented

in [31] is invoked to represent the signal in the graph domain. Since the normalized and quantized FLOCS, namely $\bar{S}_x^\varepsilon(f)$, of $S_x^\varepsilon(f)$ is symmetric about the zero spectral frequency and the zero cyclic frequency simultaneously. Therefore, we need to consider only one quadrant of $\bar{S}_x^\varepsilon(f)$. For each cyclic frequency $\varepsilon_d, d \in \{0, 1, \dots, N\}$, we can construct a corresponding graph $\mathcal{G}_{\varepsilon_d}$ according to the rules described in [31]:

- All non-negative spectral frequencies within the selected quadrant of $\bar{S}_x^\varepsilon(f)$ are regarded as the “vertices” of $\mathcal{G}_{\varepsilon_d}$, and the vertex set $\mathcal{V}_{\varepsilon_d} \stackrel{\text{def}}{=} \{f_v : v = 0, 1, \dots, N'/2\}$ can be acquired thereby.
- The amplitude difference between two adjacent vertices is taken as an “edge,” and the edge set $E_{\varepsilon_d} \stackrel{\text{def}}{=} \{\Delta \bar{S}_x^{\varepsilon_d}(v_1, v_2) : v_1, v_2 = 0, 1, \dots, N'/2\}$ can be formed as

$$\Delta \bar{S}_x^{\varepsilon_d}(v_1, v_2) \stackrel{\text{def}}{=} \begin{cases} \bar{S}_x^{\varepsilon_d}(f_{v_2}) - \bar{S}_x^{\varepsilon_d}(f_{v_1}), & \text{for } v_1 = 0, 1, \dots, N'/2 - 1, v_2 = v_1 + 1, \\ \bar{S}_x^{\varepsilon_d}(f_{v_2}) - \bar{S}_x^{\varepsilon_d}(f_{v_1}), & \text{for } v_1 = N'/2, v_2 = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

Thus, $N + 1$ graphs (one for each focused cyclic frequency ε_d) $\mathcal{G}_{\varepsilon_d} \stackrel{\text{def}}{=} (\mathcal{V}_{\varepsilon_d}, E_{\varepsilon_d})$, $d = 0, 1, \dots, N$ are established. It is obvious that $\mathcal{G}_{\varepsilon_d}$ is either a *cycle* or a *null graph* (containing no edges). Consequently, the corresponding adjacency matrix $\mathcal{A}_{\varepsilon_d}$ can be easily constructed for each $\mathcal{G}_{\varepsilon_d}$, where $\mathcal{A}_{\varepsilon_d}$ is a $(N'/2 + 1) \times (N'/2 + 1)$ square matrix.

C. Feature Extraction for Training Data

During the training stage, for the noise-free training signal of the k^{th} modulation type $\mathcal{K}_k$, one can construct a set of graphs $\mathbb{G}_k \stackrel{\text{def}}{=} \{\mathcal{G}_{\varepsilon_0}^k, \mathcal{G}_{\varepsilon_1}^k, \dots, \mathcal{G}_{\varepsilon_N}^k\}$ from its FLOCS, and the corresponding set of adjacency matrices is denoted by $\mathbb{A}_k \stackrel{\text{def}}{=} \{\mathcal{A}_{\varepsilon_0}^k, \mathcal{A}_{\varepsilon_1}^k, \dots, \mathcal{A}_{\varepsilon_N}^k\}$. It should be noticed that the graphs corresponding to all cyclic frequencies $\varepsilon_d, d = 0, 1, \dots, N$, are collected in $\mathbb{G}_k$, including the null graph(s). Once the graph set $\mathbb{G}_k$ for the training data of the k^{th} modulation type is obtained, one can extract the modulation features from its corresponding set of adjacency matrices $\mathbb{A}_k$ using the method proposed by [32].

Collect the entries from all adjacency matrices (for all cyclic frequencies) in $\mathbb{A}_k, k = 1, 2, \dots, K$ with the entry index $\iota = 1, 2, \dots, \mathcal{I}$, and write the “aggregate set of entry-indices” $\mathbb{I}$ as

$$\mathbb{I} \stackrel{\text{def}}{=} \{1, 2, \dots, \mathcal{I}\}, \quad (14)$$

where $\mathcal{I} \stackrel{\text{def}}{=} (N + 1) \cdot (N'/2 + 1)^2$. Here, one can construct K adjacency-matrix entry sets, each of which is denoted by

$$\mathbb{Z}_k \stackrel{\text{def}}{=} \{z^k(1), z^k(2), \dots, z^k(\mathcal{I})\}, \quad \text{for } k = 1, 2, \dots, K, \quad (15)$$

and $z^k(\iota), \iota \in \mathbb{I}$, is the collected entries from the adjacency matrices within $\mathbb{A}_k$ for the k^{th} modulation type.

In practice, the training signal data are divided into M segments to carry out M independent trails. For the m^{th} trial, the ι^{th} entry for the k^{th} training modulation type is denoted by $z_m^k(\iota)$, where $m = 1, 2, \dots, M$ and $k = 1, 2, \dots, K$, and the random variables for $z_m^k(\iota)$'s are denoted by $z_{k,\iota}$'s. The corresponding *probability density functions* (PDFs) to the absolute values of $z_{k,\iota}$'s are denoted by $\mathcal{P}'_{k,\iota}(|z_{k,\iota}|)$'s, which can be found in [32]. Then, the *principal modulation-type index* with the maximum summed absolute-value of the entry indexed by ι is denoted by κ_ι such that

$$\kappa_\iota \stackrel{\text{def}}{=} \arg \max_{k=1,2,\dots,K} \sum_{m=1}^M |z_m^k(\iota)|, \quad \text{for } \iota = 1, 2, \dots, \mathcal{I}. \quad (16)$$

As in [32], define the *joint Kullback-Leibler divergence* ψ_ι for the entry position ι by

$$\psi_\iota \stackrel{\text{def}}{=} \frac{1}{M} \sum_{m=1}^M \sum_{k=1, k \neq \kappa_\iota}^K \left\{ \log \left[\mathcal{P}'_{\kappa_\iota, \iota}(|z_m^{\kappa_\iota}(\iota)|) \right] - \log \left[\mathcal{P}'_{k, \iota}(|z_m^k(\iota)|) \right] \right\}. \quad (17)$$

Thus, for the entire modulation candidate set $\mathbb{K}$, the principal modulation type index set Λ and the corresponding joint Kullback-Leibler divergence set Ψ can be written as

$$\Lambda \stackrel{\text{def}}{=} \{\kappa_1, \kappa_2, \dots, \kappa_\iota, \dots, \kappa_{\mathcal{I}}\}, \quad (18)$$

$$\Psi \stackrel{\text{def}}{=} \{\psi_1, \psi_2, \dots, \psi_\iota, \dots, \psi_{\mathcal{I}}\}. \quad (19)$$

Then, according to Λ and Ψ , one can iteratively select L entry indices from $\mathbb{I}$ to construct a modulation-feature sequence $\mathbb{I}_k^{\text{training}}$ for the k^{th} modulation candidate $\mathcal{K}_k \in \mathbb{K}$ using the algorithm demonstrated in Algorithm 1 below. When the modulation-feature sequences $\mathbb{I}_k^{\text{training}}, k = 1, 2, \dots, K$ are generated, the unselected entry-indices in $\mathbb{I}$ are preserved and denoted by $\mathbb{I}_{\text{remains}}$ for further use.

Unlike the time-consuming feature-selection by human manipulation proposed in [31], the automatic feature-selection approach in Algorithm 1 not only absolutely avoids human involvement, but also can be easily extended to any arbitrary modulation candidate set systematically for its *remarkable scalability*. When the modulation candidate set is expanded or shrunk, only the features for the modulation candidate(s) which are added to or removed from the original modulation candidate set need to be generated, and the huge computational load for “re-computing the features” of all training data over the new entire modulation candidate set can be avoided thereupon. Let's discuss about these the two situations for the changes in the candidate set, namely *addition* (the candidate set is scaled up) and *subtraction* (the candidate set is scaled down), in the following.

1) *Feature-Set Modification When the Modulation-Candidate Set is Scaled Up*: Consider a new modulation type, say $\mathcal{K}_{K+1}$, is added to the original modulation candidate set $\mathbb{K}$. The expanded modulation candidate set is denoted by $\mathbb{K}^+ \stackrel{\text{def}}{=} \mathbb{K} \cup \{\mathcal{K}_{K+1}\} = \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_k, \dots, \mathcal{K}_K, \mathcal{K}_{K+1}\}, k = 1, 2, \dots, K + 1$.

Algorithm 1: Iterative Automatic Feature-Selection for Training Data.

Require: $\mathbb{I}, \Lambda, \Psi, L$
Ensure: $\mathbb{I}_{\text{remains}}, \mathbb{I}_k^{\text{training}}$ for $k = 1, 2, \dots, K$
 $\rho \leftarrow 0$
 $\zeta \leftarrow 0$
 $\mathbb{I}_k^{\text{training}} \leftarrow \emptyset, k = 1, 2, \dots, K$
 $\mathbb{I}_{\text{remains}} \leftarrow \mathbb{I}$
while $\zeta < KL$ and $\rho \leq \mathcal{I}$ **do**
 $\iota_{\rho}^{\text{opt}} \leftarrow \arg \max_{\iota \in \mathbb{I}} \psi_{\iota}.$
if $\kappa_{\rho} = k, k \in \{1, 2, \dots, K\}$ **then**
if $|\mathbb{I}_k^{\text{training}}| < L$ **then**
 $\mathbb{I}_k^{\text{training}} \leftarrow \mathbb{I}_k^{\text{training}} \cup \{\iota_{\rho}^{\text{opt}}\}$
 $\mathbb{I}_{\text{remains}} \leftarrow \mathbb{I}_{\text{remains}} \setminus \{\iota_{\rho}^{\text{opt}}\}$
 $\zeta \leftarrow \zeta + 1$
end if
end if
 $\rho \leftarrow \rho + 1$
end while
 {where $|\cdot|$ is the *get size* operator, and $\setminus$ is the *set difference* operator.}

For the new modulation type $\mathcal{K}_{K+1}$, one can also build a set of graphs $\mathbb{G}_{K+1} \stackrel{\text{def}}{=} \{\mathcal{G}_{\varepsilon_0}^{K+1}, \mathcal{G}_{\varepsilon_1}^{K+1}, \dots, \mathcal{G}_{\varepsilon_N}^{K+1}\}$ from its training data using the method mentioned in Section III-B, and the corresponding set of adjacency matrices is $\mathbb{A}_{K+1} \stackrel{\text{def}}{=} \{\mathcal{A}_{\varepsilon_0}^{K+1}, \mathcal{A}_{\varepsilon_1}^{K+1}, \dots, \mathcal{A}_{\varepsilon_N}^{K+1}\}$. Then, the entries with the indices defined by $\mathbb{I}_{\text{remains}}$ in $\mathbb{A}_{K+1}$ are exploited to construct the modulation-feature sequence $\mathbb{I}_{K+1}^{\text{training}}$. Here, the original unselected index set $\mathbb{I}_{\text{remains}}$ can be rewritten by

$$\mathbb{C}_R \stackrel{\text{def}}{=} \mathbb{I}_{\text{remains}} = \{c_1, c_2, \dots, c_w, \dots, c_W\}, \quad (20)$$

where $W \stackrel{\text{def}}{=} \mathcal{I} - KL$.

Assume that M trails are carried out. The principal modulation-type index with the maximum summed absolute-value of the entry indexed by c_w can be determined as

$$\kappa_{c_w} \stackrel{\text{def}}{=} \arg \max_{\hat{k}=1,2,\dots,K+1} \sum_{m=1}^M |z_m^{\hat{k}}(c_w)|, \quad \text{for } w = 1, 2, \dots, W. \quad (21)$$

The corresponding joint Kullback-Leibler divergence ψ_{c_w} is given by

$$\psi_{c_w} \stackrel{\text{def}}{=} \frac{1}{M} \sum_{m=1}^M \sum_{\hat{k}=1, \hat{k} \neq \kappa_{c_w}}^{K+1} \left\{ \log \left[\mathcal{P}'_{\kappa_{c_w}, c_w} \left(|z_m^{\hat{k}}(c_w)| \right) \right] - \log \left[\mathcal{P}'_{\kappa_{c_w}, c_w} \left(|z_m^{\kappa_{c_w}}(c_w)| \right) \right] \right\}. \quad (22)$$

For the entire new modulation candidate set $\mathbb{K}^+$ (including the addition), the new principal modulation-type index set Λ_R and the corresponding joint Kullback-Leibler divergence set Ψ_R can

Algorithm 2: Iterative Automatic Feature-Selection for the Modulation Candidate Set Scaled Up.

Require: $\mathbb{C}_R, \Lambda_R, \Psi_R, L, \mathbb{K}^+, \mathbb{I}_{\text{remains}}$
Ensure: $\mathbb{I}_{K+1}^{\text{training}}, \mathbb{I}_{\text{remains}}^+$
 $\rho \leftarrow 0$
 $\zeta \leftarrow 0$
 $\mathbb{I}_{K+1}^{\text{training}} \leftarrow \emptyset$
 $\mathbb{I}_{\text{remains}}^+ \leftarrow \mathbb{I}_{\text{remains}}$
while $\zeta < L$ and $\rho \leq W$ **do**
 $c_{\rho}^{\text{opt}} \leftarrow \arg \max_{c_w \in \mathbb{C}_R} \psi_{c_w}.$
if $\kappa_{\rho} = (K+1)$ **then**
if $|\mathbb{I}_{K+1}^{\text{training}}| < L$ **then**
 $\mathbb{I}_{K+1}^{\text{training}} \leftarrow \mathbb{I}_{K+1}^{\text{training}} \cup \{c_{\rho}^{\text{opt}}\}$
 $\mathbb{I}_{\text{remains}}^+ \leftarrow \mathbb{I}_{\text{remains}}^+ \setminus \{c_{\rho}^{\text{opt}}\}$
 $\zeta \leftarrow \zeta + 1$
end if
end if
 $\rho \leftarrow \rho + 1$
end while
 {where $|\cdot|$ is the *get size* operator, and $\setminus$ is the *set difference* operator.}

be written as

$$\Lambda_R \stackrel{\text{def}}{=} \{\kappa_{c_1}, \kappa_{c_2}, \dots, \kappa_{c_w}, \dots, \kappa_{c_W}\}, \quad (23)$$

$$\Psi_R \stackrel{\text{def}}{=} \{\psi_{c_1}, \psi_{c_2}, \dots, \psi_{c_w}, \dots, \psi_{c_W}\}. \quad (24)$$

Thus, the modulation-feature sequence $\mathbb{I}_{K+1}^{\text{training}}$ for the newly added modulation candidate $\mathcal{K}_{K+1} \in \mathbb{K}^+$ can be systematically constructed using the algorithm demonstrated in Algorithm 2. Here, $\mathbb{I}_{\text{remains}}^+$ stands for the unselected index set for the new expanded modulation candidate set $\mathbb{K}^+$.

2) *Feature-Set Modification When the Modulation-Candidate Set is Scaled Down:* Consider a selected modulation candidate $\mathcal{K}_{\varsigma} \in \mathbb{K}$, $\varsigma \in \{1, 2, \dots, K\}$ is removed from the original modulation candidate set $\mathbb{K}$. The shrunk modulation candidate set is thus denoted by $\mathbb{K}^- \stackrel{\text{def}}{=} \mathbb{K} \setminus \{\mathcal{K}_{\varsigma}\}$. To construct the feature sequences for the new modulation candidate set $\mathbb{K}^-$, it only needs to directly remove the existing modulation-feature sequence $\mathbb{I}_{\varsigma}^{\text{training}}$ and retain the other modulation-feature sequences $\mathbb{I}_k^{\text{training}}, k \in \{1, 2, \dots, K\} \setminus \{\varsigma\}$. No extra computations are necessary to be performed here. Consequently, we have

$$\mathbb{I}_{\text{remains}}^- = \mathbb{I}_{\varsigma}^{\text{training}} \cup \mathbb{I}_{\text{remains}}, \quad (25)$$

where

$$\mathbb{I}_{\text{remains}} \stackrel{\text{def}}{=} \mathbb{I} \setminus \left(\bigcup_{k=1}^K \mathbb{I}_k^{\text{training}} \right). \quad (26)$$

Here, $\mathbb{I}_{\text{remains}}^-$ denotes the unselected index set for the new shrunk modulation candidate set $\mathbb{K}^-$.

D. Feature Extraction for Test Data

For the test signal data, one can establish a set of graphs $\mathcal{G}$ from its FLOCS using the approach mentioned in Section III-B, and $\mathbb{A}$ indicates the corresponding adjacency matrix set. The modulation-feature sequences for the test signal data can be extracted from $\mathbb{A}$ according to $\mathbb{I}_k^{\text{training}}$, $k = 1, 2, \dots, K$. For the k^{th} modulation type $\mathcal{K}_k \in \mathbb{K}$, sort the indices $\mathbb{I}_k^{\text{training}} \rightarrow \mathbb{I}_k^{\text{test}}$ according to the magnitudes of the corresponding entries in the adjacency matrices in $\mathbb{A}$ resulting from the test data.

E. Graph-Domain Classification

Once the modulation-feature sequences $\mathbb{I}_k^{\text{test}}$, $k = 1, 2, \dots, K$ for test signal data are constructed, the *normalized Hamming distance* (NHD) $\bar{\mathcal{H}}_k$ between the feature sequence from the k^{th} training modulation type and that from the test data can be calculated as

$$\bar{\mathcal{H}}_k(\mathbb{I}_k^{\text{training}}, \mathbb{I}_k^{\text{test}}) \stackrel{\text{def}}{=} \frac{\mathcal{H}_k(\mathbb{I}_k^{\text{training}}, \mathbb{I}_k^{\text{test}})}{L}, \quad (27)$$

where $\mathcal{H}_k(\mathbb{I}_k^{\text{training}}, \mathbb{I}_k^{\text{test}})$ denotes the Hamming distance between $\mathbb{I}_k^{\text{training}}$ and $\mathbb{I}_k^{\text{test}}$. The modulation type of the test data is identified as the modulation candidate resulting in the smallest NHD.

IV. THEORETICAL PERFORMANCE ANALYSIS OF OUR PROPOSED AMC CLASSIFIER

In this section, the performance of our proposed AMC technique is theoretically studied in detail. For the k^{th} modulation type, the statistical distribution for its feature sequence $\mathbb{I}_k^{\text{training}}$, which is extracted from the training signal data using our proposed algorithm discussed in Section III, can be first established. For the test signal data regarded as the j^{th} modulation type in $\mathbb{K}$, $j \in \{1, 2, \dots, K\}$, the probabilities of NHDs between the modulation-feature sequence $\mathbb{I}_j^{\text{test}}$ and all feature sequences $\mathbb{I}_k^{\text{training}}$, $k = 1, 2, \dots, K$, can be derived according to the corresponding statistical distributions. Finally, the correct classification probability P_{cc} over the entire candidate set $\mathbb{K}$ is formulated according to the attained probabilities of NHDs. The “scalability,” namely the capability of computationally-efficient updates of P_{cc} for the occurrence of additions/removals in the modulation candidate set, is investigated.

A. Statistical Characterization

For the k^{th} modulation type in $\mathbb{K}$, the feature sequence for the training data can be rewritten as

$$\mathbb{I}_k^{\text{training}} \stackrel{\text{def}}{=} \{i_1^k, i_2^k, \dots, i_\ell^k, \dots, i_L^k\}, \quad (28)$$

where i_ℓ^k denotes the ℓ^{th} entry index for the k^{th} modulation type, $\ell = 1, 2, \dots, L$, and L is the length of the feature sequence such that $L = |\mathbb{I}_k^{\text{training}}|$. In the m^{th} trial, the i_ℓ^k -th entry for the k^{th} training modulation type is denoted by $z_m^k(i_\ell)$, where $m = 1, 2, \dots, M$ and $k = 1, 2, \dots, K$. Thus, the random variables for $z_m^k(i_\ell)$'s are denoted by z_{k,i_ℓ} 's. The corresponding PDF of z_{k,i_ℓ} can be estimated from its histogram. Define an

L -dimensional random column-vector by

$$\mathbf{Z}_k \stackrel{\text{def}}{=} [z_{k,i_1}, z_{k,i_2}, \dots, z_{k,i_\ell}, \dots, z_{k,i_L}]^T. \quad (29)$$

In order to simplify the derivation, we assume that the L -dimensional random vector $\mathbf{Z}_k$ complies with a joint Gaussian distribution or a *multivariate normal* (MVN) distribution. Then, this hypothesis will be examined using the Shapiro-Wilk test subject to the p -value [36]. If the p -value is less than the chosen significance level, say 0.05, the null hypothesis is rejected and it is evident that the test data do not constitute an MVN distribution. In this situation, the Edgeworth series is invoked instead to approximate the PDF of the random variables z_{k,i_ℓ} in terms of its cumulants.

1) *Joint Gaussian Distribution*: If $\mathbf{Z}_k$ can be justified to constitute an MVN distribution through the Shapiro-Wilk test, the corresponding joint PDF can be expressed by

$$\mathcal{P}(\mathbf{Z}_k) \stackrel{\text{def}}{=} \frac{1}{\sqrt{(2\pi)^L |\boldsymbol{\Sigma}|}} \times \exp \left[-\frac{1}{2} (\mathbf{Z}_k - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{Z}_k - \boldsymbol{\mu}) \right], \quad (30)$$

where $\boldsymbol{\mu} \in \mathbb{R}^L$ is the L -dimensional mean vector of $\mathbf{Z}_k$; $\boldsymbol{\Sigma} \in \mathbb{R}^{L \times L}$ is the covariance matrix of $\mathbf{Z}_k$ which is symmetric and positive definite. The *cumulative distribution function* (CDF) of $\mathbf{Z}_k$ can be obtained by integrating the joint PDF $\mathcal{P}(\mathbf{Z}_k)$ over the given region $\mathfrak{R}$ that is a hyperrectangle such that

$$\Phi_L(\mathbf{b}, \mathbf{d}; \boldsymbol{\Sigma}, \boldsymbol{\mu}) \stackrel{\text{def}}{=} \frac{1}{\sqrt{(2\pi)^L |\boldsymbol{\Sigma}|}} \int_{\mathfrak{B}_1}^{\mathfrak{D}_1} \int_{\mathfrak{B}_2}^{\mathfrak{D}_2} \dots \int_{\mathfrak{B}_L}^{\mathfrak{D}_L} \times \exp \left[-\frac{1}{2} (\mathbf{Z}_k - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{Z}_k - \boldsymbol{\mu}) \right] dz_{k,i_1} \dots dz_{k,i_L}, \quad (31)$$

where

$$\mathfrak{R} \stackrel{\text{def}}{=} [\mathfrak{B}, \mathfrak{D}] \stackrel{\text{def}}{=} \begin{bmatrix} \mathfrak{B}_1 \\ \mathfrak{D}_1 \end{bmatrix} \times \begin{bmatrix} \mathfrak{B}_2 \\ \mathfrak{D}_2 \end{bmatrix} \times \dots \times \begin{bmatrix} \mathfrak{B}_L \\ \mathfrak{D}_L \end{bmatrix}, \quad -\infty \leq \mathfrak{B}_\ell \leq \mathfrak{D}_\ell \leq +\infty, \quad \ell = 1, 2, \dots, L. \quad (32)$$

Since there is no closed form of $\Phi_L(\mathfrak{B}, \mathfrak{D}; \boldsymbol{\Sigma}, \boldsymbol{\mu})$, the *Quasi-Monte Carlo* (QMC) method can be employed to approximate the value of CDF [37].

2) *Non-Gaussian Distribution*: If $\mathbf{Z}_k$ does not pass the Shapiro-Wilk test, which means $\mathbf{Z}_k$ does not comply with the MVN Distribution, the Edgeworth series is carried out instead to approximate its PDF based on the corresponding cumulants. Consider the random variables z_{k,i_ℓ} 's. The estimated PDFs of z_{k,i_ℓ} 's can be constructed in terms of the first three cumulants [38] such that

$$\mathcal{P}(z_{k,i_\ell}) \stackrel{\text{def}}{=} \left[1 + \frac{\lambda_3 H_3(z_{k,i_\ell})}{6} + \frac{\lambda_4 H_4(z_{k,i_\ell})}{24} \right] \phi(z_{k,i_\ell}), \quad (33)$$

with

$$H_3(z_{k,i_\ell}) \stackrel{\text{def}}{=} z_{k,i_\ell}^3 - 3z_{k,i_\ell}, \quad (34)$$

$$H_4(z_{k,i_\ell}) \stackrel{\text{def}}{=} z_{k,i_\ell}^4 - 6z_{k,i_\ell}^2 + 3, \quad (35)$$

$$\lambda_3 \stackrel{\text{def}}{=} E \left\{ [z_{k,i_\ell} - E(z_{k,i_\ell})]^3 \right\}, \quad (36)$$

$$\lambda_4 \stackrel{\text{def}}{=} E \left\{ [z_{k,i_\ell} - E(z_{k,i_\ell})]^4 \right\} - 3 \left\{ E(z_{k,i_\ell}^2) - E[(z_{k,i_\ell})^2] \right\}^2, \quad (37)$$

where λ_3 denotes the *skewness* of z_{k,i_ℓ} which is an “asymmetry” measure of the PDF; λ_4 is the *kurtosis* of z_{k,i_ℓ} which measures the “tailedness” of the PDF; $H_3(x)$ and $H_4(x)$ are the second- and third-order *normalized Hermite polynomials*, respectively; $\phi(\xi)$ is the *perturbation* of a Gaussian PDF and expressed by

$$\phi(\xi) \stackrel{\text{def}}{=} \frac{1}{\sqrt{2\pi}} e^{-\frac{\xi^2}{2}}. \quad (38)$$

In order to reduce the computational complexity, we assume that the L -dimensional random vector $\mathbf{Z}_k$ consists of statistically independent components. Thus the corresponding PDF to $\mathbf{Z}_k$ is given by

$$\mathcal{P}(\mathbf{Z}_k) = \prod_{l=1}^L \mathcal{P}(z_{k,i_l}), \quad (39)$$

where $\mathcal{P}(z_{k,i_\ell})$ represents the marginal PDF of z_{k,i_ℓ} estimated by Eq. (33).

B. Probability of NHD

For the given test signal data of the j^{th} modulation type $\mathcal{K}_j$ in $\mathbb{K}$, an L -dimensional random vector $\mathbb{I}_j^{\text{test}}$, $j \in \{1, 2, \dots, K\}$ will be constructed from the test signal data according to the training feature sequence $\mathbb{I}_k^{\text{training}}$, $k \in \{1, 2, \dots, K\}$ using the approach mentioned in Section III-D. Then, the K test modulation-feature sequences can be obtained and rewritten by $\mathbb{I}_{j|k}^{\text{test}}$, $k = 1, 2, \dots, K$. One can define a random vector $\tilde{\mathbf{Z}}_{j|k} \stackrel{\text{def}}{=} [\tilde{z}_{j|k,i_1}, \tilde{z}_{j|k,i_2}, \dots, \tilde{z}_{j|k,i_L}]^T$ for the test signal data of the j^{th} modulation type under the k^{th} modulation type assumption, where $\tilde{z}_{j|k,i_\ell}$ denotes the random variable corresponding to the entry indexed by $i_\ell^k \in \mathbb{I}_k^{\text{training}}$ in the adjacency matrix set $\mathbb{A}$ resulting from the test signal of the j^{th} modulation type. The PDF of $\tilde{\mathbf{Z}}_{j|k}$ can be easily formed via our proposed method in Section IV-A.

Since the NHDs measure the distinctions between the feature sequences derived from a test signal and the training signals of all modulation types, and the modulation type is determined by the training modulation model which results in the smallest NHD for the test data. Consider all modulation candidates $\mathcal{K}_k$ in $\mathbb{K}$, $k = 1, 2, \dots, K$, the probability of the test signal for the j^{th} modulation type to result in the feature sequences $\mathbb{I}_k^{\text{training}}$ for the k^{th} modulation type (without any error) is denoted by

$P(\mathcal{K}_j|\mathcal{K}_k : 0)$ such that

$$P(\mathcal{K}_j|\mathcal{K}_k : 0) = \int_{-\infty}^{\infty} \int_{-\infty}^{\tilde{z}_{j|k,i_1}} \int_{-\infty}^{\tilde{z}_{j|k,i_2}} \dots \int_{-\infty}^{\tilde{z}_{j|k,i_L-1}} \mathcal{P}(\tilde{\mathbf{Z}}_{j|k}) \times d\tilde{z}_{j|k,i_1} d\tilde{z}_{j|k,i_2} \dots d\tilde{z}_{j|k,i_L}, \quad (40)$$

where $\mathcal{P}(\tilde{\mathbf{Z}}_{j|k})$ denotes the PDF of $\tilde{\mathbf{Z}}_{j|k}$. Similarly, the probability of the test signal for the j^{th} modulation type to result in the feature sequences $\mathbb{I}_k^{\text{training}}$ for the k^{th} modulation type (with a single error) is denoted by $P(\mathcal{K}_j|\mathcal{K}_k : 1)$ such that

$$P(\mathcal{K}_j|\mathcal{K}_k : 1) = \int_{-\infty}^{\infty} \int_{-\infty}^{\tilde{z}_{j|k,i_1}} \int_{-\infty}^{\tilde{z}_{j|k,i_2}} \dots \int_{-\infty}^{\tilde{z}_{j|k,i_L-1}} \mathcal{P}(\tilde{\mathbf{Z}}_{j|k}) \times d\tilde{z}_{j|k,i_1} d\tilde{z}_{j|k,i_2} \dots d\tilde{z}_{j|k,i_L} + \int_{-\infty}^{\infty} \int_{-\infty}^{\tilde{z}_{j|k,i_1}} \int_{-\infty}^{\tilde{z}_{j|k,i_2}} \dots \int_{-\infty}^{\tilde{z}_{j|k,i_L-1}} \mathcal{P}(\tilde{\mathbf{Z}}_{j|k}) \times d\tilde{z}_{j|k,i_1} d\tilde{z}_{j|k,i_2} \dots d\tilde{z}_{j|k,i_L} + \dots \quad (41)$$

Consequently, one can sequentially derive all probabilities of the test signal for the j^{th} modulation type to result in the feature sequences $\mathbb{I}_k^{\text{training}}$ for the k^{th} modulation type with v errors, namely $P(\mathcal{K}_j|\mathcal{K}_k : v)$, for $v = 0, 1, \dots, L$.

C. P_{cc} Formulation

Once all of the probabilities $P(\mathcal{K}_j|\mathcal{K}_k : v)$ are derived, the correct classification probability P_{cc} over the modulation candidate set $\mathbb{K}$ can be calculated by

$$P_{cc}(K) \stackrel{\text{def}}{=} \sum_{k=1}^K P(\mathcal{K}_k) \sum_{v=0}^{L-1} P(\mathcal{K}_k|\mathcal{K}_k : v) \times \prod_{j=1}^{K, j \neq k} \left[1 - \frac{\left\lceil \frac{v \|\mathbb{I}_j^{\text{training}}\|}{L} \right\rceil}{\sum_{q=0}^L P(\mathcal{K}_k|\mathcal{K}_j : q)} \right], \quad (42)$$

where $P(\mathcal{K}_k)$ is the *a priori* probabilities of the occurrence of the modulation $\mathcal{K}_k$, $k = 1, 2, \dots, K$.

D. The Scalability of P_{cc} Formulation

Consider the iterative modulation-feature-selection algorithms stated in Section III-C. It is apparent from Eq. (42) that the P_{cc} formulation can be easily extended to any arbitrary modulation candidate set systematically. When the modulation candidate set is changed, P_{cc} for the new modulation candidate set can be modified iteratively without exhaustive effort simply by adding or subtracting a term to/from the previous version of P_{cc} . Since only one term resulting from the added/subtracted modulation type needs to be calculated, the computational complexity for the complete update of P_{cc} is thus significantly reduced.

1) *Iterative P_{cc} Formulation for the Up-Scaled Candidate Set:* When a new additional modulation type $\mathcal{K}_{K+1}$ is involved, the size of the modulation candidate set is expanded from K to $K+1$ and the new modulation candidate set is therefore denoted by $\mathbb{K}^+ \stackrel{\text{def}}{=} \mathbb{K} \cup \{\mathcal{K}_{K+1}\} \stackrel{\text{def}}{=} \{\mathcal{K}_{\hat{k}} | \hat{k} = 1, 2, \dots, K+1\}$. Here, the index of the new joined modulation type $\mathcal{K}_{K+1}$ in $\mathbb{K}^+$ is set to $\hat{k} = K+1$, and the indices of the original modulation types in $\mathbb{K}$ are retained in $\mathbb{K}^+$.

The training modulation-feature sequence $\mathbb{I}_{K+1}^{\text{training}}$ for the new modulation type $\mathcal{K}_{K+1}$ is first systematically constructed using the method introduced in Section III-C1. Then, the PDF $\mathcal{P}(\mathbf{Z}_{K+1})$ of the random vector $\mathbf{Z}_{K+1}$ corresponding to $\mathbb{I}_{K+1}^{\text{training}}$ is established. Furthermore, the probabilities of the test signal for the j^{th} modulation type to result in the feature sequences $\mathbb{I}_{K+1}^{\text{training}}$ for the $(K+1)^{\text{th}}$ modulation type with v errors, namely $P(\mathcal{K}_j | \mathcal{K}_{K+1} : v)$'s, $v = 1, 2, \dots, L$ can be derived according to Eqs. (40) and (41) in Section IV-B. Once the $(K+1)L$ probabilities $P(\mathcal{K}_j | \mathcal{K}_{K+1} : v)$'s are obtained, the correct classification probability $P_{cc}(K+1)$ over the new modulation candidate set $\mathbb{K}^+$ can be easily determined according to the previous version of $P_{cc}(K)$ for the original modulation candidate set $\mathbb{K}$. It is given by

$$P_{cc}(K+1) = \left(\frac{K}{K+1} \right) P_{cc}(K) + \Delta P_{cc}(K+1), \quad (43)$$

where

$$\Delta P_{cc}(K+1) \stackrel{\text{def}}{=} P(\mathcal{K}_{K+1}) \sum_{v=0}^{L-1} P(\mathcal{K}_{K+1} | \mathcal{K}_{K+1} : v) \times \prod_{j=1}^K \left[1 - \sum_{q=0}^{\left\lfloor \frac{v |\mathbb{I}_{K+1}^{\text{training}}|}{L} \right\rfloor} P(\mathcal{K}_{K+1} | \mathcal{K}_j : q) \right]. \quad (44)$$

Consider another existing AMC algorithm in [31]; when the size of the modulation candidate set is expanded from K to $K+1$, $P_{cc}(K+1)$ for the new modulation candidate set $\mathbb{K}^+$ must be exhaustively recalculated, and the computational complexity is $\mathcal{O}[(K+1)^2]$. On the contrary, from Eqs. (43) and (44), it is obvious that $P_{cc}(K+1)$ for the new modulation candidate set $\mathbb{K}^+$ can be easily obtained by adding a correction term $\Delta P_{cc}(K+1)$ resulting from the extra modulation candidate $\mathcal{K}_{K+1}$ to the previous version of $P_{cc}(K)$ for the original modulation candidate set $\mathbb{K}$. Hence, only the correction term $\Delta P_{cc}(K+1)$ needs to be calculated, and the incurred extra computational complexity is remarkably reduced to $\mathcal{O}(2K+1)$.

2) *Iterative P_{cc} Formulation for the Down-Scaled Candidate Set:* When one modulation candidate $\mathcal{K}_{\varsigma} \in \mathbb{K}$, $\varsigma \in \{1, 2, \dots, K\}$ is removed from the original modulation candidate set $\mathbb{K}$, the shrunk modulation candidate set is denoted by $\mathbb{K}^- \stackrel{\text{def}}{=} \mathbb{K} \setminus \{\mathcal{K}_{\varsigma}\}$. The correct classification probability $P_{cc}(K-1)$ over the new modulation candidate set $\mathbb{K}^-$ can be easily derived according to $P_{cc}(K)$ of the original modulation

candidate set $\mathbb{K}$. It is given by

$$P_{cc}(K-1) = \left(\frac{K}{K-1} \right) P_{cc}(K) - \nabla P_{cc}(K-1), \quad (45)$$

where

$$\nabla P_{cc}(K-1) \stackrel{\text{def}}{=} P(\mathcal{K}_{\varsigma}) \sum_{v=0}^{L-1} P(\mathcal{K}_{\varsigma} | \mathcal{K}_{\varsigma} : v) \times \prod_{j=1}^{K, j \neq \varsigma} \left[1 - \sum_{q=0}^{\left\lfloor \frac{v |\mathbb{I}_{K+1}^{\text{training}}|}{L} \right\rfloor} P(\mathcal{K}_{\varsigma} | \mathcal{K}_j : q) \right]. \quad (46)$$

It is obvious that the contribution $\nabla P_{cc}(K-1)$ generated by the removed modulation type $\mathcal{K}_{\varsigma}$ in $\mathbb{K}$ is directly subtracted from the original version of $P_{cc}(K)$. No extra computations need to be performed in this scenario.

V. SIMULATION

In this section, the effectiveness of our proposed AMC approach using the graph-based FLOCS analysis is first validated via Monte Carlo simulations. Then, the performance of our proposed AMC approach is evaluated, and the sensitivity of our scheme to the characteristics of α -stable noise is also investigated. Finally, we justify the robustness of our derived iterative theoretical P_{cc} formulation. The same modulation candidate set as in [31], [32], which consists of *binary phase-shift keying* (BPSK), *quadrature phase-shift keying* (QPSK), *offset quadrature phase-shift keying* (OQPSK), *binary frequency-shift keying* (2FSK), *quadrature frequency-shift keying* (4FSK), and *minimum shift keying* (MSK), is adopted here.

A. Effectiveness of Our Proposed Method

The effectiveness of our proposed AMC technique in the α -stable noise background is first compared with the existing AMC_G method proposed in [31], [32]. The correct classification probability P_{cc} over the entire candidate set $\mathbb{K}$ and the individual *correct recognition percentages* (CRPs) versus MSNR for $\mathcal{K}_k$, $k = 1, 2, \dots, K$ are evaluated via Monte Carlo simulations. The FFT window size for generating FLOCS is set to be 32. The MSNR η ranges from -10 to 20 dB subject to the additive α -stable noise, and the characteristic exponent is set to $\alpha = 1.5$. For every MSNR condition, one thousand Monte Carlo trials are carried out by the computer. In each trial, 8,192 samples (symbols) are randomly generated for every individual modulation type. The number of quantization levels is 16.

The correct recognition percentages for the aforementioned modulation candidate set are depicted in Fig. 4. It is obvious from Fig. 4 that the correct recognition percentages resulting from our proposed scheme (denoted by “FLOCS” in the figure) for all modulation types in $\mathbb{K}$ can converge to 100% when $\eta \geq 4$ dB. On the other hand, the existing AMC_G method (denoted by “SCS” in the figure) based on the conventional cyclic spectrum analysis always leads to poor performance for $\eta \leq 20$

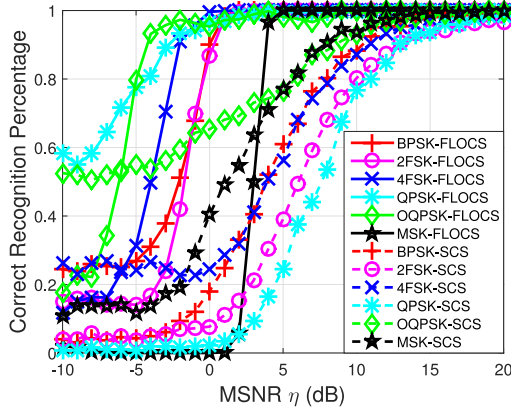


Fig. 4. Correct recognition percentages versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

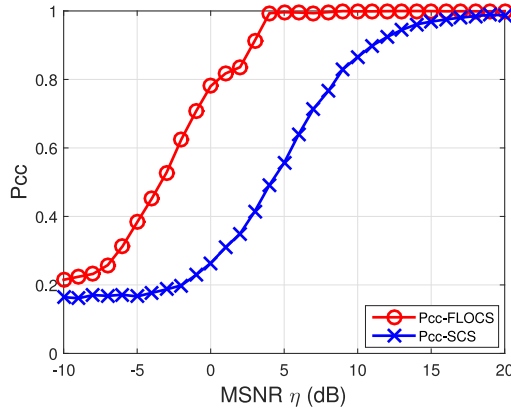


Fig. 5. Probabilities of correct classification P_{cc} versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

dB. Meanwhile, the probabilities of correct classification P_{cc} for the given six modulation candidates in $\mathbb{K}$ in the presence of α -stable noise are also delineated in Fig. 5. According to Fig. 5, P_{cc} generated by the existing AMC_G method (denoted by “SCS” in the figure) cannot lead to any satisfactory performance when $\eta \leq 20$ dB. On the contrary, the perfect performance ($P_{cc} = 100\%$) can be attained by use of our proposed new AMC technique when the MSNR η approaches 4 dB. According to the above-mentioned simulation results, our proposed AMC scheme using the graph-based FLOCS analysis can effectively identify the correct modulation type of the signal corrupted by α -stable noise even for low MSNRs.

B. Performance Analysis of Our Proposed AMC Technique

In order to further assess the performance of our proposed AMC technique, the existing AMC technique based on the human-selected features of FLOCS is also investigated for comparison. Since the FLOCS analysis can result in the polyspectra of the signal degraded by α -stable noise, the second-order cyclic spectrum analysis upon which the conventional AMC approaches depend can be considered its special case under the Gaussian noise assumption. Here, four (human-selected) distinct

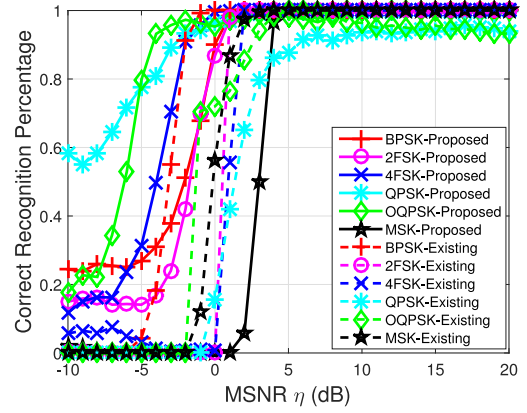


Fig. 6. Correct recognition percentages versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

features defined in [7]–[9] are extracted from the FLOCS and utilized to identify the modulation scheme for the additive α -stable noise background. These four “human-selected” features consist of the number of δ impulses of $S_x^\varepsilon(f)$, the maximum normalized decline value of $S_x^{2f_c}(f)$, the ratio of the maximum response of $\{S_x^\varepsilon(f_c)|\varepsilon > 0\}$ to the maximum response of $\{S_x^\varepsilon(0)\}$, and the spectral frequency index of the maximum value of $\{S_x^\varepsilon(f)|\varepsilon > 0\}$, where f_c is the carrier frequency. Then, P_{cc} for the given candidate set $\mathbb{K}$ and the individual CRPs versus MSNR for each $\mathcal{K}_k$, $k = 1, 2, \dots, K$ resulting from our proposed AMC method (using the systematically selected optimal features by computer itself) are evaluated and compared with those resulting from the above-mentioned human-selected features of FLOCS. The same simulation set-up presented in Section V-A is also adopted here.

The correct recognition percentages for the given modulation candidate set are demonstrated in Fig. 6. According to Fig. 6, it is apparent that the CRPs resulting from our proposed scheme (denoted by “Proposed” in the figure) for QPSK, OQPSK, 2FSK, and 4FSK can achieve 100% accuracy even in the low MSNR conditions. The CRPs for the existing AMC method using the human-selected features of FLOCS (denoted by “Existing” in the figure) are also plotted in Fig. 6. The minimum MSNR required by our proposed AMC scheme to guarantee perfect recognition for BPSK and MSK is 2 dB while that required by the counterpart is yet much larger. Moreover, the CRPs resulting from the human-selected features of FLOCS for QPSK and OQPSK cannot reach up to 100% even for high MSNRs.

The probabilities of correct classification P_{cc} for the aforementioned six modulation candidates in the presence of α -stable noise are depicted in Fig. 7. It is obvious that P_{cc} resulting from the human-selected features of FLOCS for the entire modulation candidate set cannot reach 100% even for high MSNRs. On the contrary, our proposed AMC scheme can lead to around 80% when the MSNR η is as low as 0 dB and reach up to 100% (the perfect performance) when the MSNR η approaches 4 dB. In summary, a remarkable performance improvement has been observed by use of our proposed AMC approach over the existing method.

Furthermore, the impact of the characteristic exponent α associated with the α -stable noise on the performance of our

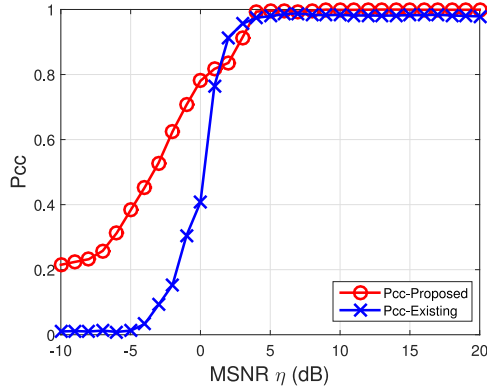


Fig. 7. Probabilities of correct classification P_{cc} versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

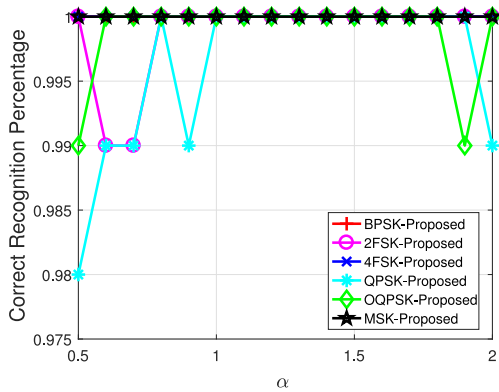


Fig. 8. Correct recognition percentages versus α for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

proposed AMC technique is also investigated here. For a given MSNR $\eta = 10$ dB, the correct recognition percentages versus α for the entire modulation candidate set $\mathbb{K}$ are illustrated in Fig. 8.

It is conspicuous from Fig. 8 that for all modulation candidates in $\mathbb{K}$, our proposed AMC scheme can achieve the perfect performance when α falls within the range $[1, 1.8]$. The CRPs for QPSK, 2FSK, and OQPSK slightly decline as $\alpha < 1$, but these tiny deteriorations could still be tolerable.

C. Performance Comparison Between Our Proposed AMC Technique and the LB Approach

The performance of the SSD based ML classification method proposed in [26] is evaluated in the presence of α -stable noise via simulation and compared with our proposed AMC technique. The LB method in [26] invokes the SSD to extract the deterministic modulated signal component from the received signal corrupted by the α -stable noise, and then the ML based classifier is employed for AMC. The parameters λ , α , and f_c specified in [26] are set to 0.01, 1.5, and 2, respectively. The modulation candidate set and other system parameters specified in Section V still remain unchanged here. The correct recognition percentages for the given modulation candidate set are depicted in Fig. 9.

It is obvious from Fig. 9 that the SSD based ML classifier (denoted by “Likelihood” in the figure) performs poorly for

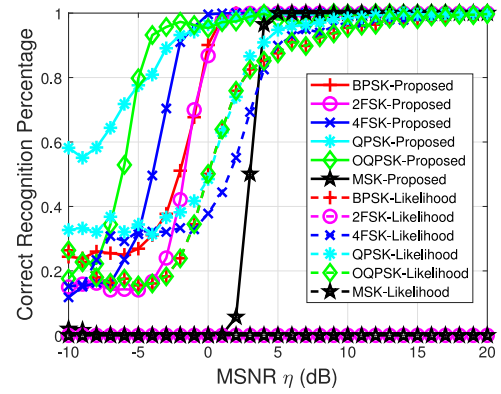


Fig. 9. Correct recognition percentages versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} resulting from our proposed AMC technique and the existing LB classifier in the presence of α -stable noise.

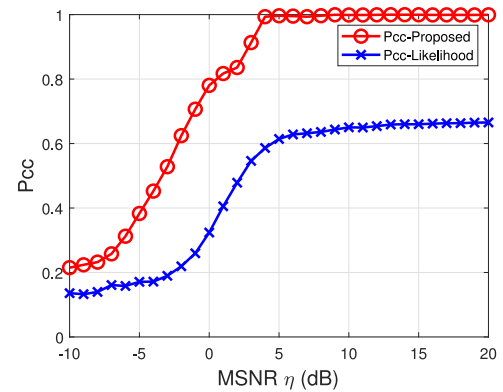


Fig. 10. Probabilities of correct classification P_{cc} versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} resulting from our proposed AMC technique and the existing LB classifier in the presence of α -stable noise.

2FSK and MSK for most MSNR values. For classifying BPSK, 4FSK, QPSK, and OQPSK signals, our proposed AMC method (denoted by “Proposed” in the figure) can lead to the perfect accuracy as $\eta \geq 2$ dB. However, the existing algorithm requires the MSNR to be at least 10 dB for achieving the same performance. Meanwhile, the probabilities of correct classification P_{cc} for the aforementioned six modulation candidates in the presence of α -stable noise are delineated in Fig. 10. According to Fig. 10, the P_{cc} of the SSD based ML classifier can only reach up to 60% when $\eta \geq 5$ dB. On the other hand, our proposed method can lead to the perfect accuracy, ($P_{cc} = 100\%$), under the same MSNR condition. In general, our proposed AMC method outperforms the existing LB classification method for almost all MSNR conditions.

D. Computational Complexity Analysis of the Proposed AMC Technique

The computational complexities of the proposed new AMC technique and the existing method mentioned stated in Section V-B are theoretically analyzed here. According to the description of the proposed AMC method in Section III and the existing approach proposed in [7]–[9], it is obvious that these two methods both utilize the FLOCS for AMC. Meanwhile, for

TABLE I
THE COMPUTATIONAL COMPLEXITIES OF THE PROPOSED AMC ALGORITHM
AND THE EXISTING AMC METHOD IN [7]–[9]

Operation	The proposed method	The existing method
Addition	$(N+1)\left(\frac{N'}{2}+1\right)^2 + LK$	$\left(\frac{3N'}{2}+3\right)$
Comparison	$K-1$	$\left\lceil \frac{N'}{2}(N+5) + 3N-2 \right\rceil + (K-1)$
Division	K	1
Quick Sort	$K \cdot O(L^2)$	0

the proposed new AMC technique, since the training signals are clean ideal signals, their feature sets can be calculated by both algorithms in advance and stored in the memory. Hence, we only need to consider the computational complexity of the feature extraction for testing signals and the classification. The computational complexities of the proposed new AMC technique and the existing method proposed in [7]–[9] in the worst case are listed in Table I.

Form Table I, it is apparent that four operations are involved in the proposed new AMC technique, and the total computational complexity is mainly contributed by additions and quick sorts. On the contrary, there involve three operations in the existing method proposed in [7]–[9], where comparisons contribute to the majority portion of the total computational complexity. Because the time complexities of these operations highly depend on particular computer architectures, we utilize simulations to benchmark them here. The system parameters still remain unchanged as before. The total time complexities of the proposed new AMC algorithm and the existing approach are 4.643 s and 0.312 s, respectively. Hence, we have to admit that the computational complexity of our proposed AMC technique is larger than that of the existing method in the worst scenario.

On the other hand, note that the adjacency matrices of the graphs built from FLOCS are all sparse matrices, where the non-zero entries can only be found along the sub-diagonal just above the main diagonal for each adjacency matrix. Hence, we only need to extract the feature sequence from these non-zero entries, and the number of additions in the proposed new AMC algorithm can be dramatically reduced to $(N+1)\left(\frac{N'}{2}+1\right) + LK$. Thus, the total computational complexity of our proposed AMC technique can further decrease, while the total time complexity of the proposed AMC algorithm is reduced to 0.2733 s. Considering the performances of the proposed new AMC method depicted in Fig. 6 and Fig. 7, the proposed new AMC technique can lead to a superior performance to the existing method proposed in [7]–[9] subject to about the same computational complexity.

E. Robustness of Theoretical P_{cc} Formulation

According to the discussion in Section IV-C, the probabilities of correct classification P_{cc} for the given modulation candidate set $\mathbb{K}$ in the presence of α -stable noise can be theoretically formulated using Eq. (42). Our derived P_{cc} formula, which effectively avoids tedious simulations and computations, provides an efficient and convenient tool for the *prompt performance*

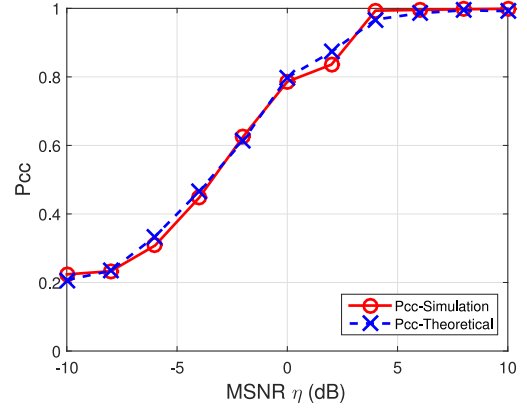


Fig. 11. Comparison of P_{cc} 's versus MSNR for {BPSK, QPSK, OQPSK, 2FSK, 4FSK, MSK} in the presence of α -stable noise.

assessment of our proposed AMC scheme. The accuracy of such a formula of P_{cc} is benchmarked by comparing the simulation results of P_{cc} from Monte Carlo simulations with the theoretical results by simply enumerating Eq. (42). Here, the same modulation candidate set $\mathbb{K}$, which includes BPSK, QPSK, OQPSK, 2FSK, 4FSK, and MSK, is adopted, and the simulation set-up still remains unchanged as before.

The comparison of P_{cc} 's for all candidates in $\mathbb{K}$ in the presence of α -stable noise is demonstrated in Fig. 11. It is apparent from Fig. 11 that the theoretical results of P_{cc} using Eq. (42) (denoted by “Theoretical” in the figure) and P_{cc} 's generated by Monte Carlo simulations (denoted by “Simulation” in the figure) match each other closely. The root-mean-square error (RMSE) of P_{cc} is just 0.0186 or 1.86%. Therefore, according to our derived formula of P_{cc} given by Eq. (42), once the graph feature set is iteratively constructed, the classification accuracy P_{cc} for a given modulation candidate set can be directly calculated from a theoretical perspective with high precision. The performance evaluation does not have to rely on cumbersome simulations and computations. It should be noticed that the optimal iterative feature-selection mechanism proposed in this work can make the graph-based AMC approach *fully scalable*, and P_{cc} can also be iteratively and efficiently computed according to Eqs. (43)–(46) as the modulation candidate set keeps being scaled up or down.

VI. CONCLUSION

An innovative AMC scheme using the graph-based FLOCS analysis in the presence of α -stable noise is presented. This graph-based AMC mechanism is systematically established on the graph representations in the FLOCS domain to identify the modulation scheme according to the discrepancies between the graph features iteratively derived from the training and test data. Based on this new framework, the highly efficient, iterative graph-feature extraction algorithm is devised and the correct classification probability P_{cc} over a given modulation candidate set is theoretically formulated. The scalability and efficiency of our proposed paradigm are elaborately demonstrated by the derivation of the iterative modulation-feature update rule and the theoretical formula of P_{cc} , which not only is very robust but

also avoids the incurred time-consuming simulations and computations from the resized modulation candidate sets. Numerical results are also presented to exhibit the outstanding performance of our proposed new AMC scheme over the additive α -stable noise.

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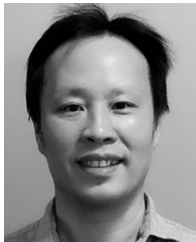
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